

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

EDMS 12-401-3

SPECIFICATION

FOR

STEP VOLTAGE REGULATORS

FOR MV NETWORKS UP TO 24 kV

Issue: July-2025/ Rev- 3

- This revision contains option items that must be selected by EDC before bidding.
- EDMS 25-401 for Mineral Insulating Oil shall be attached.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

CONTENTS

CLAUSE	DESCRIPTION	PAGE
-	GLOSSARY	3
1	SCOPE	5
2	APPLICABLE STANDARDS	5
3	ENVIRONMENTAL CONDITIONS	6
4	DESIGN CRITERIA	6
5	CONSTRUCTION REQUIREMENTS	10
6	CONTROL PANNEL	16
7	VOLTAGE REGULATOR EFFICIENCY	17
8	MARKING	18
9	TESTING and INSPECTION	19
10	SUBMITTALS	21
11	GUARANTEE	23
12	PACKING and SHIPMENT	23
13	TRAINING and CUSTOMER SUPPORT	23
14	TECHNICAL DATA SCHEDULE	24
-	APPENDIX A: VOLTAGE REGULATOR CONTROL	28

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

GLOSSARY

EEHC	Egyptian Electricity Holding Company
EDMS	Egyptian Distribution Material Specifications
EDC	Electricity Distribution Company
EHVRC	Extra-High Voltage Research Center
SVR	Step-type Voltage Regulator
MV	Medium Voltage
ONAN	Oil Natural Air Natural cooling
ONAF	Oil Natural Air Forced cooling
RMS	Root Mean Square value
BIL	Basic Insulation Level
AC	Alternating Current
DC	Direct Current
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
ANSI	American National Standard Institute
SI	International System of units
OLTC	On-Load Tap Changer
ACSR	Aluminum Conductor Steel Reinforced
AAAC	All Aluminum Alloy Conductor
MOV	Metal Oxide Varistor (also called as Zinc-oxide arrester)
SCADA	Supervisory Control and Data Acquisition
IDPT	Internal Differential Potential Transformer

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

CT	Current Transformer
VT	Voltage Transformer
IP	Ingress Protection
LED	Light-Emitting Diode
LCD	Liquid Crystal Display
CPU	Central Processing Unit
USB	Universal Serial Bus
RFI	Radio Frequency Interference
SC	Short Circuit
PS	Power Supply
RBAC	Role-Based Access Control
THD	Total Harmonic Distortion
HMI	Human-Machine Interface

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

1. SCOPE

This specification covers the minimum technical requirements for design, manufacturing, testing, inspection, supply and delivery of outdoor, single-phase, oil-immersed step-type voltage regulators (SVR). The required voltage regulators are intended to regulate the medium voltage level (11 kV or 22 kV, 50 Hz) of the power networks of Egyptian Electricity Distribution Companies (EDCs).

2. APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

Standard No.	Description
IEC 60076-21 / IEEE Std C57.15	Power Transformers - Part 21 Standard requirements, terminology, and test code for step-voltage regulators
IEC 60076-7	Power Transformers - Part 7 Loading guide for oil-immersed power transformers
IEEE Std C57.91	IEEE Guide for loading mineral-oil-immersed transformers and step-voltage regulators
IEC 60076-20	Power Transformers - Part 20 Energy efficiency
IEC 60214-1	Tap-changers - Part 1: Performance requirements and test methods
IEC/IEEE 60214-2	Tap-changers - Part 2: Application guidelines
IEEE Std C57.131	IEEE Standard requirements for tap-changers
IEC 60137	Bushings for alternating voltages above 1 kV.
IEC 60296	Fluids for electrotechnical applications – Mineral insulating oils for electrical equipment
IEC 60099-4	Surge arresters - Part 4: Metal-oxide surge arresters without gaps for a.c. systems

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

IEEE Std C57.13	IEEE Standard requirements for instrument transformers
IEEE Std C57.12.31	IEEE Standard for pole-mounted equipment - Enclosure integrity
IEEE Std C57.19.00	IEEE Standard general requirements and test procedure for outdoor power apparatus bushings
IEC 60255 Measuring relays and protection equipment	Part 1: Common requirements
	Part 21-1: Vibration, shock, bump and seismic tests on measuring relays and protection equipment - Vibration tests (sinusoidal)
	Part 26: Electromagnetic compatibility requirements
	Part 27: Product safety requirements

3. ENVIRONMENTAL CONDITIONS

The electrical and mechanical properties of the required voltage regulators shall be guaranteed under the environmental conditions given in Table (2).

Table (2)

Ambient temperature	-5°C to +45°C (50°C is an option according to _ _ EDC requirements), (35°C monthly average of the hottest month, 25°C yearly average)
Maximum relative humidity	95 %
Altitude	Up to 1000 m above mean sea-level

4. DESIGN CRITERIA

4.1 Ratings

- Based on 50 Hz operating frequency, the maximum continuous current/power ratings of SVR is selectable from the values given in Table 3.
- The voltage regulator shall be able to deliver the maximum continuous current/power rating without exceeding the allowable temperature rise limit. Forced air (ONAF) cooling is not acceptable to achieve the nominal current/power rating.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- The SVR design (Type A or Type B) is determined by the manufacture provided that the selected design shall meets all the requirements of this specification.
- A single-phase SVR must be completely self-contained and provide $\pm 10\%$ regulation in thirty-two (32) steps of approximately 5/8% each.
- A bank of three single-phase SVRs shall be suitable to be grouped in closed-delta connection to provide overall voltage regulation $\pm 15\%$ of the three-phase system. Two single-phase SVRs shall be suitable to be grouped in open-delta connection to provide overall voltage regulation $\pm 10\%$ of the three-phase system.
- Regulators of the voltage ratings 22 kV shall have internal and external tap settings for operation at the standard regulated voltages of 11 kV.

Table (3) Ratings for oil-immersed 50 Hz SVRs (single-phase)

Maximum system voltage (U_m) kV	Nominal system voltage kV	Voltage Regulator Ratings		Maximum no-load (excitation) sound pressure levels dB
		kVA	A	
12	11	55	50	51
		110	100	55
		165	150	55
		220	200	55
		330	300	60
		440	400	60
		550	500	61
		660	600	61
24	22	110	50	55
		220	100	55
		330	150	60
		440	200	60
		660	300	61
		880	400	62

- Voltage regulators, including their controls, shall be suitable for operation within the voltage limits given in Table 4, provided that the rated load current is not exceeded.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

Table (4) Operating voltage limits

Nominal system voltage kV	Voltage supply ratio	Input voltage (kV)			Output voltage (kV)	
		Minimum	Maximum at rated-load amperes	Maximum at no-load	Minimum	Maximum at rated-load amperes or at no-load
11	91.7	8.96	11.55	12.1	9.49	12.1
22	183.3	17.92	23.1	24.2	18.975	24.2

- Voltage regulators shall permit additional current-carrying capabilities at reduced regulation, as shown in Table 5. The adjustment of this supplementary continuous-current ratings shall be located inside the position-indicator faceplate to prevent inadvertent adjustment. In addition, a soft-type adjustment of this feature shall be available through the control keypad.

Table (5) Supplementary continuous-current ratings

Range of voltage regulation (%)	Continuous-current rating (%)
10.0	100
8.75	110
7.5	120
6.25	135
5.0	160

- Permissible overload capacity shall be in accordance to IEEE Std C57.91 and IEC 60076-7.

4.2 Short-circuit Withstand

- Step-voltage regulators shall be designed and constructed to withstand the mechanical and thermal stresses produced by external short-circuits at a maximum value of 25 times the base RMS symmetrical rated load current to a maximum

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

requirement of 16 kA. The short-circuit, thermal withstand capability, shall be based on a symmetrical current with a duration of 2 s.

- The first-cycle asymmetrical peak current the voltage regulator is required to withstand shall be determined by:

$$I_{sc} \text{ (peak asym)} = k \times I_{sc} \text{ (RMS sym)}$$

where $k = 2.19$ for regulators < 250 kVA

$k = 2.55$ for regulators ≥ 250 kVA

4.3 Temperature Rise Limits

- The SVR shall be able to deliver its maximum continuous ratings without exceeding the following permissible temperature rise limits:
 - Top oil level temperature rise 55 °C
 - Winding temperature rise measured by resistance method 55 °C
 - Hottest-spot winding temperature-rise 65 °C

4.4 Insulation Level

- Voltage regulators shall be designed to provide coordinated applied-voltage and lightning impulse insulation levels on line terminals, and applied-voltage insulation levels on neutral terminals. The identity of a set of coordinated levels shall be its basic impulse insulation level (BIL), as shown in Table 6.

Table (6) Dielectric insulation levels for SVRs

Nominal system voltage kV	BIL kV	Applied-voltage insulation level (kV RMS)	Impulse levels		
			Full wave	Chopped wave	
			kV crest	kV crest	Minimum time to flashover (μs)
11	95	34	95	105	1.8
22	150	50	150	165	3

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

4.5 Sound Pressure Level

- The sound pressure level produced by the SVR under a no-load (excitation) condition shall not exceed the values specified in Table 3.

5. CONSTRUCTION REQUIREMENTS

- The SVRs shall be designed such that they can be partially or completely untanked for inspection and maintenance without disconnecting any internal electrical or mechanical connections.

5.1 Tap-changing Mechanism

- The tap-changing mechanism shall be completely oil-immersed.
- An electrical feedback circuit shall be incorporated with the tap-changer motor circuit and control to ensure accurate indication of tap position and number of operations. Accuracy is to be maintained for all dielectric fluids used with the voltage regulator.
- Regulator control system shall have Class 1 accuracy for all ratings of load voltage regulation by the use of load side voltage transformers or utility windings. Control calculations, using tap position, are not acceptable for regulation of the normal load-side voltage.
- The regulator shall be supplied with a center-tapped reactor for the purpose of maintaining continuity while the tap-changer is changing position. The reactor shall be isolated from ground to provide protection from lightning and switching surges. Construction of the reactor shall minimize no-load losses and leakage current.
- Internal secondary wiring shall be color coded or labeled for easy identification.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

5.2 Windings

- The conductor material of the main coil shall be copper for shunt winding and aluminum for series windings. Series winding made of copper conductor is an option according to _ _ EDC requirement.
- The regulator main coil, reactor, and potential transformer shall include thermally upgraded insulation to permit operation up to 65 °C rise without loss of life to the insulating system. At 65 °C rise, the regulator shall provide 12% extra current capacity over the base current rating. A suitably patterned, epoxy-coated insulation paper shall be used in all windings. Prior to assembly of the main core and coil assembly, the windings are to be baked with sufficient mechanical pressure exerted on the sides of the coil to maximize a complete bonding of the insulation to maintain required short-circuit current withstand.
- The temperature of the conductor material in the windings of voltage regulators under the short-circuit condition shall not exceed 250 °C for a copper conductor or 200 °C for an electrical conductor (EC) aluminum.

5.3 Cooling

- Cooling is done by the natural circulation of the internal insulating oil and the external air (i.e. ONAN cooling). The SVRs shall be suitable for use with mineral insulating oils that meet the requirements of IEC 60296 and EDMS 25-401.
- Voltage regulators shall be supplied with initial filling.

5.4 Tank

- Voltage regulators shall have a sealed-tank, liquid-preservation system. Sealed-tank construction is a construction in which the interior of the tank is sealed to prevent the introduction of external atmosphere into the tank. The voltage regulator shall remain effectively sealed for a top liquid temperature range of -20 °C to +110 °C for

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

continuous operation at rated power. The top cover shall be bolted to the main tank.

- The completely assembled enclosure shall be of sufficient strength to withstand an internal pressure of 49 kPa (7 psig) without permanent deformation.
- The tank can be made of stainless steel sheet or galvanized steel sheet or any equivalent material with sufficient thickness to withstand the expected mechanical forces. The tank and its accessories shall be adequately protected against corrosion and the supplier shall include a statement of protection proposed.
- The external parts of the tank and control enclosure shall be painted Light Gray number 70, Munsell notation 5BG7.0/0.4. Finishing of the external parts of the tank and control enclosure shall meet the coating and security requirements of IEEE Std C57.12.31 standard. The inside of the tank and bottom of the cover shall also be primed and/or painted to avoid oil contamination.
- All bolts, studs, nuts and washers shall have corrosion proof locking features. They shall confirm to BS-3692 and shall be plated to BS-7371-12.

5.5 Pressure-relief Valve

- As a part of the normal operation, the tank shall be provided with a replaceable pressure-relief valve to relieve excess pressure due to normal temperature variation of top liquid and/or due to on-load tap-changer operations. The pressure-relief device vents at approximately 27.5 kPa (4 psig).
- The replaceable pressure-relief valve shall be located on the tank above the 110 °C top liquid level. The valve shall be located so it does not interfere with the use of support lugs and lifting lugs. It shall not be located in the quadrant of the tank that contains the control enclosure.
- Exposed parts shall be of weather- and corrosion-resistant materials.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- The valve shall have a pull ring for manually reducing pressure to atmospheric level using a standard hook stick, and shall be capable of withstanding a static pull force of 112 N for 1 min without permanent deformation. The valve shall withstand for 1 min a static force of 445 N applied normal to its longitudinal axis at the outermost extremity of the body.

5.6 Step Position Indicator

- An external, corrosion-resistant step position indicator with upper and lower drag hands shall indicate the tap-changer position. The position indicator shall be polymer constructed, mounted above the oil level, and slanted downward at a 45-degree angle for ease of reading from ground level.

5.7 Bushings

- The voltage regulator shall be fitted with Porcelain bushings conforming to latest relevant international standards IEC 60137 and IEC 60815. The bushings shall be compatible with the service conditions and system parameters of _ _ EDC. The bushings shall be terminated by flat pad terminals to accept vertical connection with overhead line conductors (ACSR or AAAC) of cross-sectional area starting from 70 mm² up to 300 mm².
- The BIL rating of the bushings shall be compatible with the BIL of the regulator. The bushing designations (S, L, and SL) shall be permanently marked on the regulator cover adjacent to the bushings. The S, L, and SL bushings must be interchangeable with each other.
- Bushings for use in SVRs shall have the minimum creep distance and insulation levels as listed in Table 7. The color of the bushings shall conform to Light Gray Number 70, Munsell Notation 5BG7.0/0.4, in accordance to IEEE Std C57.12.31.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

Table (7) Electrical characteristics of SVR bushings

Nominal system voltage kV	BIL kV	Minimum Creep distance mm	Applied-voltage withstand		Impulse full-wave dry withstand kV crest (1,2 × 50 μs)
			1 min dry (kV RMS)	10 s wet (kV RMS)	
11	95	480	35	30	95
22	150	960	60	50	150

5.8 Lightning Arresters

- Each single-phase SVR shall be provided with an external Metal Oxide Varistor (MOV-type) bypass arrester connected across the series winding between the source (S) and load bushings (L). For units rated 11 kV, the series arrester shall be rated 3 kV. For units rated 22 kV, the series arrester shall be rated 6 kV.
- Each single-phase SVR shall be provided with Stainless steel mounting bosses for the addition of three shunt surge arresters adjacent to the source (S), load (L), and source-load (SL) bushings. The bosses shall be fully welded around their circumference. Spot welding is not acceptable.
- Each single-phase SVR shall be supplied complete with three zinc-oxide shunt arresters that meet the requirements of IEC 60099-4 and EDMS 11-400.

5.9 Accessories and Fittings

- The supplied SVR shall be equipped with the following accessories and fittings:
 - Temperature gauge shall be dial type with re-settable maximum indicator. Scale shall be from 0 °C to 120 °C.
 - External oil level indicator shall be a gauge that indicates oil level and color.
 - Draining valve equipped with a non-detachable handle and protecting plug shall be located at the bottom of the tank. A filling pipe shall be located above

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

the full oil level, at the top of the tank. This pipe shall have internal threads and be fitted with a brass bung and gasket.

- A copper solderless bolted connector for tank grounding which will accommodate conductor up to 70 mm².
 - Lifting lugs shall be designed so that the Regulator can be lifted without damage to itself or to its fittings. All necessary fittings for mounting shall be supplied.
 - A voltage limit control shall be provided that prevents the Regulator from further raising, its output above, or going below the preset values of the limit control once the limit is reached.
 - An external electrical connection between the cover and tank shall be supplied to allow the cover-suspended internal assembly and tank to be grounded together to eliminate voltage differentials during energizing.
- All regulators shall be provided with a base suitable for securing them to an elevating structure. All regulators shall have two stainless steel ground pads located diagonally opposite from each other. Each pad shall have two stainless steel 1/2"-13 UNC ground provisions. All grounding provisions are to be located near the base of the regulator.
 - A multi-conductor 600 V, -50 °C to 105 °C cable with disconnect plugs at each end shall provide the connection between the internal circuitry of the voltage regulator and the control.
 - An electronic device shall be provided to protect the internal CT from high voltages due to the control cable being disconnected or cut while the voltage regulator is energized.
 - The motor starting capacitor shall be located inside the control enclosure and be

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

easily accessible for maintenance.

- The control cable connecting between the control cabinet and voltage regulator shall be of a quick-connect type which allows for removal and replacement of the control enclosure while the voltage regulator is energized. Hardwired control cables are not acceptable.

6. CONTROL PANNEL

- Each single phase unit of SVR shall have one control unit in a weather resistant enclosure with provision for locking.
- The control panel shall have a degree of protection IP-54 as per IEC 60529.
- The control panel shall be constructed as a single unit and be capable of being removed from the enclosure without disconnecting components or shall be capable of being disconnected from the control cable.
- Conductors of control cable shall be color coded or labeled for easy identification.
- Electronic components shall be hermetically sealed, identified, reached easily for testing and capable of operating at the specified ambient temperature.
- The control panel shall be fully wired and ready for service and shall meet all the requirements detailed in Appendix A.
- A minimum five (5) meter of control cable shall be provided between the regulator and control box. The cable and controls shall be factory fitted to the regulator. The location of any plug on the cable shall be identified, but should preferably be at the control panel.
- SVRs shall be supplied complete with the necessary VTs and CTs for control and power purposes.
- Inter-locking shall be provided to prevent energizing the high voltage terminals, in the event of an external 120 V supply being applied to the control panel.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

7. VOLTAGE REGULATOR EFFICIENCY

- The SVR shall be designed for maximum efficiency and minimum losses. Offers of SVRs will be “Loss Evaluated” in that the cost of losses will factor into the purchasing decision. Offers shall be evaluated according to IEC 60076-20 by the following capitalization formula:

$$TOC = BP + A \times L_{NL} + B \times L_{LD}$$

Where TOC = Total capitalized cost in Egyptian Pounds (EGP)

BP = Initial cost (Bidding Price) of regulator in EGP.

A -factor represents the present value of the cost of no-load losses around the regulator lifetime (20 years). B -factor represents the present value of the cost of load losses around the regulator lifetime.

$A = 110 \times C$ EGP/Watt of no-load losses. $B = 36 \times C$ EGP/Watt of load losses.

C is the valorization of the energy in EGP/kWh according to the maximum commercial cost of energy as per EEHC latest update.

L_{NL} is the average No-Load Losses in Watts

L_{LD} is the average Load Losses in Watts

- Guaranteed no-load (core) and load (winding) losses of SVR shall be provided during bidding. No-load losses at Neutral and 1R positions and load losses at 16R, 15R, 15L and 16L positions shall be provided. These losses shall be corrected to a reference temperature of 75 °C.
- No-load and Load losses submitted in the tender shall be treated as guaranteed values. Any increase in these values at the time of testing shall not be accepted.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

8. MARKING

- Each Voltage Regulator shall be provided with an easily readable two nameplates of corrosion resistance material (Laser-Etched). One nameplate is mounted on the regulator tank and the other is mounted on the front of control cabinet.
- Each nameplate will include at a minimum the following:
 - a) manufacturer's name;
 - b) type and form designation or the equivalent;
 - c) cooling class;
 - d) serial number;
 - e) month and year of manufacturing (not coded);
 - f) number of phases;
 - g) rated kVA;
 - h) rated current;
 - i) supplementary continuous-current ratings;
 - j) rated voltage
 - k) voltage transformer ratio;
 - l) rated range of regulation;
 - m) rated frequency;
 - n) impulse level, full wave in kilovolts (kV);
 - o) untanking weight;
 - p) total weight;
 - q) insulating liquid type;
 - r) volume of insulating liquid (weight of insulating liquid when specified in place of volume);
 - s) conductor material;
 - t) average winding rise in degrees Celsius (°C);
 - u) Complete diagrams showing the leads and internal connections and their markings, including polarity markings, and the voltage obtainable with the various connections;
 - v) installation and operating instructions reference;
 - w) symmetrical short-circuit withstand ampere rating with time duration;

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- x) asymmetrical short-circuit withstand, first peak ampere rating (when specified);
- y) on-load tap-changer model number and maximum through-current rating;
- z) tap-changer motor capacitor rating;
- aa) Purchase order Number (P.O. Number).

9. TESTING and INSPECTION

9.1 General

- Tests shall be carried out in accordance with the latest relevant standard and as specified herein.
- All tests shall be carried out by and at the expense of the tenderer who shall supply all the apparatus, instruments and equipment.
- Routine tests shall be carried out in the presence and under the control of the inspecting engineer of _ _ EDC, except in specified or approved cases where test certificates are accepted. All test results shall be reviewed and approved in writing by _ _ EDC.
- The contractor should advise testing dates. The contractor shall not be entitled to any extension of the time of completion due to the failure of any test or the rejection of any part of the materials as a result of any test.
- Routine tests shall be carried out in the supplier's factory or at any other mutually approved place.

9.2 Type Tests

- The following type tests shall be made on representative voltage regulators to substantiate the ratings assigned to all other voltage regulators of basically the same design:

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- | | | |
|----|-------------------------|--|
| 1- | Temperature rise test. | (Clause 9.10 of IEC 60076-21 and IEEE C57.15) |
| 2- | Lightning impulse test. | (Clause 9.7.2 of IEC 60076-21 and IEEE C57.15) |
| 3- | Short-circuit test. | (Clause 9.11 of IEC 60076-21 and IEEE C57.15) |
| 4- | Sound level test. | (Clause 9.12 of IEC 60076-21 and IEEE C57.15) |

9.3 Routine Tests

- Routine tests shall be made in accordance to IEC 60076-21 and IEEE C57.15 on all voltage regulators per the following list:
 - a) Leak test;
 - b) Operational test of voltage regulator including all devices. Controlled devices such as on-load tap-changers, position indicators, etc., shall be operated for proper functioning;
 - c) Resistance measurements of all windings (clause 9.2);
 - d) Polarity test (clause 9.3);
 - e) Ratio test (clause 9.4);
 - f) No-load loss at rated voltage and rated frequency (clause 9.5);
 - g) Excitation current at rated voltage and rated frequency (clause 9.5);
 - h) Impedance and load loss at rated current and rated frequency (clause 9.6);
 - i) Lightning impulse test (clause 9.7.3);
 - j) Applied-voltage test (clause 9.7.4);
 - k) Induced-voltage test (clause 9.7.5);
 - l) Insulation power factor test (clause 9.7.6);
 - m) Insulation resistance test (clause 9.7.7);
 - n) On-load tap-changer tests (clause 9.8);
 - o) Control system tests (clause 9.9).

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

9.4 Rejection of Materials

- Failure of any material to comply with any one of the requirements of this specification shall constitute grounds for rejecting that material, except if the ____EDC inspection engineer convinced that the material could satisfactorily be repaired. In such case, the contractor should be obliged to replace the defective material without being entitled to any extra payment or to any extension of time to complete the contract.

9.4.1 Rejection for decreased efficiency:

- The contractor shall keep the no-load losses and load-losses to be less than or equal the guaranteed values.
- Should the measured losses exceed the guaranteed values, then ____EDC representative should reject the voltage regulator with such losses.

9.4.2 Replacement of rejected material and equipment

- In case of any rejected material or equipment, the contractor shall be obliged to replace it without extra payment.

10. SUBMITTALS

10.1 Submittals required with tender

- The tenderer shall complete and return one legible copy of the attached Data Schedule for each rate of the offered voltage regulator.
- Dimensional drawings of the offered voltage regulator including interconnection diagrams of all components and assemblies. A nameplate design of the offered voltage regulator including all of its information shall be attached.
- Original catalogues shall be submitted to facilitate evaluation of the offer.
- All information should be in English language, clearly readable, with IEC standard symbols and SI units.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- Type test reports/certificates from an independent testing laboratory shall be submitted to __ EDC.
- Routine test reports from Extra-High Voltage Research Center (EHVRC) or any other approved testing laboratory for a regulator of the same offered rate or higher.
- Approval certificate issued by Egyptian Electricity Holding Company (EEHC).
- The bidder shall separately quote the cost of one bypass/disconnect and one disconnect switch per each single phase voltage regulator, which shall be compatible with the regulator offered.
- List of deviations and clauses to which exception is taken (if any).

10.2 Submittals required following award of contract

- Details of manufacturing and test schedules.
- Factory test reports.
- The contractor should provide with shipment of the voltage regulator complete installation, operation and maintenance instruction/manuals suitable for the specific supplied item(s), including information from the manufacturer of each purchased accessory. It should include interconnected diagrams, drawings and instructions describing procedures for items that will be shipped to be assembled and interconnected for operation when installed. These should contain strict instructions as required for carrying, moving or lifting the voltage regulator and accessories.
- Key interlocking should be identified, and serial number should be supplied.
- The contractor should provide a complete set of “as manufactured” drawings and wiring diagrams including interconnection diagrams of all components and assemblies of the equipment which assist in operation and maintenance of the equipment.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

11. GUARANTEE

- The supplier guarantee the voltage regulator, control unit and all accessories against any defects arising out of faulty design or workmanship, or of defective material for a period of two (2) years from the date of delivery.
- In case of supplying for the first time, the guarantee period should be extended to 4 years at least from delivery date.

12. PACKING and SHIPMENT

- SVRs shall be delivered ready for service with the control panel secured to the tank.
- SVRs shall be packed in sea-worthy, non-returnable crates, designed to protect against mechanical damage and ingress of moisture and dust. Regulators shall not be packed in any organic material.
- Packing notes shall be included in each crate giving a description of the goods packed. Components or materials, if subjected to a shelf life limitation, shall have the final date of use prominently shown on all cases.
- Each crate shall be printed with the following information:
 - Regulator catalogue number.
 - Purchase Order number.
 - Manufacturer's name
 - Year of manufacture.
 - Gross weight.
 - Position of slinging points and other relevant handling instructions.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

13. TRAINING and CUSTOMER SUPPORT

- The supplier shall provide adequate on-site training for a sufficient number of ____ EDC representatives, to be agreed by (.....EDC) and supplier. The training should include and not limited to the following:
 - Installation and Programming.
 - Testing and Commissioning.
 - Troubleshooting.
 - Operating and Maintenance
- The Supplier must have a dedicated service center and support line in Egypt to cover voltage regulators, tap-changer, control unit and any other deliverables.
- __ EDC reserves the right to claim a free after-sales service for the entire warranty period. After-sales service should include and not limited to the following:
 - Technical support.
 - Maintenance.
 - Original Spare parts.
- All services, spare parts, materials and accessories required during the after-sales maintenance should be delivered by the supplier immediately wherever required (warehouse, site, laboratory, etc.)

14. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

TECHNICAL DATA SCHEDULE
MV STEP VOLTAGE REGULATOR (SVR)

__ EDC Inquiry No: _____ Item No: _____

No.	DESCRIPTION	UNIT	-- EDC Specified Values*		Vendor Proposed Values**
			12 kV System	24 kV System	
1	Manufacturer's name				
2	Country of Origin				
3	SVR model				
4	Type and form designation	Type A (Type B is option)			
5	The standard to which regulator manufactured	IEC 60076-21 / IEEE C57.15			
6	Cooling class	ONAN			
7	Number of phases	Single-phase			
8	Rated Voltage	kV	12	24	
9	Nominal system voltage	kV	11	22	
10	Rated Frequency	Hz	50		
11	Rated Power	kVA	As per tender requirement		
12	Rated current	A	As per tender requirement		
13	Maximum no-load (excitation) sound pressure levels	dB	As per Table 3		
14	Rated range of regulation:				
14.1	- Single-phase unit	%	10	10	
14.2	- Three units in Open delta	%	10	10	
14.3	- Three units in Closed delta	%	15	15	
15	Number of steps	32 (16L : 16R)			
16	Operating voltage limits:				
16.1	- Minimum input voltage	kV	8.96	17.92	
16.2	- Maximum input voltage at rated load current	kV	11.55	23.1	
16.3	- Maximum input voltage at no-load	kV	12.1	24.2	

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

16.4	- Minimum output voltage	kV	9.49	18.975	
16.5	- Maximum output voltage at rated load current or no-load	kV	12.1	24.2	
17	Voltage regulator losses:				
17.1	- No-load losses at Neutral position at 75 °C	Watt	Provided by supplier		
17.2	- No-load losses at 1R position at 75 °C	Watt	Provided by supplier		
17.3	- Load losses at 16R position at 75 °C	Watt	Provided by supplier		
17.4	- Load losses at 15R position at 75 °C	Watt	Provided by supplier		
17.5	- Load losses at 15L position at 75 °C	Watt	Provided by supplier		
17.6	- Load losses at 16L position at 75 °C	Watt	Provided by supplier		
18	Maximum Impedance Voltage at 16R position	%	Provided by supplier		
19	Supplementary continuous-current ratings as per Table 5?		Yes		
20	Maximum ambient temperature at which rating apply	°C	45		
21	Temperature rise limits at maximum continuous rating:				
21.1	- Maximum Top oil level temperature rise	°C	55		
21.2	- Maximum Winding temperature rise	°C	55		
21.3	- Maximum Hot-spot winding temperature-rise	°C	65		
22	Symmetrical Short circuit withstand I _{sc} (RMS sym)	kA	As per clause 4.6		
23	Duration of symmetrical short-circuit current	s	As per clause 4.6		
24	Max. Asymmetrical Short circuit I _{sc} (peak asym)	kA	As per clause 4.6		
25	Dielectric insulation levels:				
25.1	- BIL	kV	95	150	
25.2	- Applied-voltage insulation level (RMS)	kV	34	50	
25.3	- Full-wave impulse level	kV	95	150	
25.4	- Chopped-wave impulse level	kV	105	165	
26	Voltage transformer ratio	-	91.7	183.3	
27	Current transformer ratio	-	Rated current / 0.2		
28	On-load tap-changer (OLTC) model				
29	Maximum through-current rating of OLTC	A	Provided by supplier		

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

30	OLTC speed (time to move from N to 16R)	s		
31	Maximum number of life operations of OLTC	steps	Provided by supplier	
32	Tap-changer motor capacitor rating		Provided by supplier	
33	Control unit model			
34	Control unit meets all the requirements of clause 8 and Appendix A?		Yes	
35	Conductor material of shunt winding	-	Copper	
36	Conductor material of series winding	-	Copper / Aluminum	
37	Operating pressure of relief valve	kPa	27.5	
38	Minimum creepage distance of bushings	mm	480	960
39	Type of insulating liquid	-	Mineral oil	
40	Volume of insulating liquid	Liters	Provided by supplier	
41	Un-tanking weight	kg	Provided by supplier	
42	Gross weight of completely assembled regulator	kg	Provided by supplier	
43	Guarantee period from delivery date	Years	2	
44	Nameplate marking meets all the requirements?		Yes	
45	Inspection and testing meets all the requirements?		Yes	
46	Submittals meet all the requirements?		Yes	

(*) - Values to be provided / proposed by the vendor.

(**) – Please, provide explanations for deviations if any. (Use separate sheet if necessary.)

I / We guarantee the technical data given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

APPENDIX A

VOLTAGE REGULATOR CONTROL

1. Voltage Regulator Control general requirements

a. General Feature

- The voltage regulator control shall be:
 - Microprocessor-based
 - Modular in design to allow for easy installation and removal of accessory hardware to enable the addition of features and functions.
- Control LCD display: The control LCD display shall have these features:
 - A backlit LCD to display control menus, settings, metering values and other parameters.
 - LCD display shall have an adjustment for contrast or have clear vision during different operation periods.
- Motor fuse: A user-replaceable motor-fuse shall be provided. The face of the control must be clearly marked with the fuse amperage rating near the fuse holder. A spare fuse shall be included within the control enclosure.
- Heater: The control shall have a built-in heater to warm the control cabinet and reduce condensation inside the enclosure controlled by suitable hygostat if needed.
(Optional according to EDC requirements)
- Keys (Switches and/or Buttons): These switches and buttons shall be available for operation of the voltage regulator and control.
 - Full alphanumeric keypad or bottoms for quick, easy modification of control parameters and entry of security passwords.
 - Three-position voltage source switch.
 - The switch is to be labeled INTERNAL-OFF-EXTERNAL and will allow the control to be energized from the regulator's internal voltage supply or from an external source.
 - The switch must be configured to prevent inadvertent energization of the internal supply when an external power source is applied.
 - In the OFF position the control will be de-energized.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- Keys of the control mode selection.
 - The key is to be labeled AUTO/REMOTE-OFF-LOCAL MANUAL and allows for selection of option for control and operation of the tap changer.
 - In the AUTO/REMOTE position the tap changer will be automatically controlled by the control or through remote SCADA
 - In the LOCAL MANUAL position, the operation of the tap-changer will be via a key located on the front of the control.
 - In the OFF position, the tap-changer will be disabled from operation.
- A local-manual momentary three-position switch to operate the tap-changer.
 - The switch will have two momentary positions labeled RAISE and LOWER which will operate the tap-changer motor in the raise and lower direction.
 - When the switch is release, it will automatically return to the center off position.
 - The switch must be hardwired to operate the tap-changer motor the even when the control CPU is not functional.
- A position-indicator drag-hand reset button.
 - The button will be labeled DRAG HAND RESET
- Supervisory function
 - During supervisory off statues: the SCADA control of the tap changer motor and parameter configurations is inhibited.
 - During supervisory is active, SCADA will be able to continue monitoring control setting, parameters, and metering.
- Terminals: These terminals shall be available on the face of the control panel.
 - Voltmeter test terminals having the following characteristics:
 - High impedance path which enables measurement of the voltage of the load-bushing potential transformer (or control winding) secondary.
 - On board short-circuit protection components provided to inhibit an accidental back-feeding of the potential transformer (or control winding) by way of the voltmeter test terminals.
 - External source terminals which provide this functionality:
 - Provide a connection point for 120 Vac power to energize the control and operate the tap-changer motor when the POWER switch is on the internal position.
 - Other control accessories must also be energized when power is applied to the terminals. These accessories including communications interface cards, I/O modules, DC power supplies and any other of the control's modular accessories.
 - The terminals shall be prevented from creating an accidental back-feeding of the potential transformer (or control winding) by use of an external power switch interlock.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- LED Indicators: These LED indicators shall be present on the face of the control:
 - Dual neutral position LEDs that are actuated via a tap-changer neutral switch to provide neutral position indication. These LEDs must be of the high- intensity type and easily readable in direct sunlight.
 - The control shall have the following voltage regulation status Indications through (LEDs or LCD display) on the face of the control:
 - Band-edge indicators, high and low, to display when the compensated voltage falls outside of the set voltage band.
 - Voltage Limiter, high and low, to display when the load voltage falls outside of the set Voltage Limiter limits.
 - Auto Tap Block to indicate a condition in which the control is inhibited from automatic voltage regulation.
 - Reverse Power to inhibit when the control is regulating voltage in the normal reverse direction.
 - Voltage Reduction to indicate when voltage reduction is active.
 - Two alarm condition LEDs, one marked ALARM to indicate highest-level alarm conditions and a second marked WARNING to indicated second-level alarm conditions.
 - Diagnostic error LED to indicate that a control diagnostic self-test has found a test failure.
 - Two sets of Transmit and Receive LEDs to provide visual indication of SCADA communications activity through the two communications ports.
 - Two user-defined LED indicator shall be provided to display the presence of user-defined conditions. The operation of these LEDs is to be adjustable using designated interface software.
 - Data port indication for the connection of USB storage device and local SCADA activity with designated interface software.

b. Accuracy

- The control system of a voltage regulator shall have an overall system error not exceeding $\pm 1\%$. The accuracy requirement is based on the combined performance of the control device and the voltage and current sensing apparatus.
- The voltage source accuracy shall be determined on a nominal secondary voltage base of 120 V and a burden of 10 VA.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- The current source accuracy shall be determined on a nominal 0.2 A secondary current and a burden of 3.5 VA.

c. Temperature Rating

- The control shall operate in an operating environment temperature range of -25°C to $+65^{\circ}\text{C}$.
- Control device components shall withstand a temperature range -25°C to $+85^{\circ}\text{C}$ without damage or loss of control.

d. Environmental Interference

- Control shall meet all environmental, insulation coordination, electrostatic discharge, RFI and interference tests specified in the latest version of IEEE Std C57.15 standard.

e. Enclosure

- The regulator control shall be mounted in a weather-resistant enclosure, which is capable of being padlocked, absent other appropriate security options.
- The control enclosure shall have an external 1/2"–13 UNC stainless steel welded ground boss.
- The control enclosure shall have a means to secure manuals supplied for the apparatus and control on the inside of the enclosure. Storage of the manuals, in this manner, shall not adversely affect the control unit.
- The control unit and the enclosure shall be earthed.
- The enclosure IP shall be 54.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

f. Mounting

- The control panel shall be hinge-mounted and designed for easy replacement.
- The control panel shall be constructed to provide direct interchangeability without removal of the control enclosure.
- Visible means shall be provided in the back of the enclosure to de-energize the control panel and associated circuitry in the back of the enclosure, and to short the voltage regulator's internal current transformer prior to testing or removal of control.

g. Wire Leads

- All leads in the control enclosure shall be color coded or connections labeled for easy identification.

h. Terminal Boards

- The terminal boards of the control inside the control enclosure shall consist of dead-front clamp-style quick connectors for ease of access and protection of operator.

i. Control Circuit Boards

- All printed circuit boards shall be conformal coated for fungi and moisture protection.

j. Ratio-Correction Transformers

- A ratio-correction transformer for each internal voltage supply shall be provided for easy access and fine voltage adjustment.
- Ratio correction tap connections and the corresponding system voltage shall be clearly identified on unit nameplates.

k. Control Human Machine Interface Structure

- The control menu shall be displayed in easy way that allows direct access to display all setting, parameters and metering values.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- Menu System: A nested menu structure shall be utilized to enable easy and logically navigation to all settings and metering values, and most parameters.
- Metering: A system of interface screens shall be available to display compiled control settings and parameters which are frequently needed when operating a voltage regulator. The compiled information shall display:
 - The compensated voltage, voltage band range, and setting parameters used to arrive at the displayed information.
 - The load voltage, and voltage limiter limits.
 - The load current, current threshold, reverse sensing mode, and auto-blocking status.
 - The current tap position, supplementary current limits, and a display that the tap position has reached the supplementary current limits.

1. Voltage Regulation Features

- Basic voltage regulation settings: These basic voltage regulation settings shall be available:
 - Set voltage with a range of 100.0 to 135.0 volts in increments of 0.1 volts for both forward and reverse power flow.
 - Bandwidth with a range of 1.0 to 6.0 volts in increments of 0.1 volts for both forward and reverse power flow.
 - Time delay with a range of 5 to 180 seconds in 1 second increments for both forward and reverse power flow.
 - Line Drop Compensation settings for resistance and reactance with a range of adjustable of -25 to +25 volts in increments of 0.1 V both for forward and reverse power flow.
 - System configuration with preconfigured options of Wye (Star), Delta Lead and Delta Lag settings. A separate setting of phase angle for Delta connections is not acceptable.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- Four control operating modes:
 - Sequential – Calls for voltage regulation when the compensated voltage falls outside the voltage band for the time delay period. The time delay counter resets each time the compensated voltage crosses into the voltage band.
- Reverse power operation: The control reverse power capabilities shall have these features:
 - Allow automatic operation of the voltage regulator under reverse power flow conditions.
 - Establish a current sense threshold with a minimum setting of 1% of the current transformer primary rating. The current sense threshold will be the level at which the control is able to accurately determine the power flow direction.
 - Determine reverse power using these strategies:
 - Compare voltage and current phase angles when real current is above the current sense threshold.
 - These reverse sensing modes shall be available:
 - Locked Forward
 - Locked Reverse
 - Reverse Idle
 - Bidirectional
 - Neutral Idle
 - Cogeneration
 - Reactive Bidirectional
 - Bias Bidirectional
 - Reverse Cogeneration
 - Bias Cogeneration (Bias cogeneration is an option according to __ EDC requirements)
- Voltage limiter: A voltage-limiting capability providing this functionality:
 - Provide first-house protection by preventing the load voltage from exceeding a set limit when using the line-drop compensation feature.
- Voltage Reduction: A configurable voltage reduction functionality to reduce voltage from 0.1% to 10% as option.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

m. Communications

– Local Communication:

- USB data-drive port with programming and data retrieval capabilities shall be available. A user must be able to upload or download control parameters and data without requiring the use of a computer or external hand-held device.
- A front-panel data port shall be available for temporary direct communications to a computer with designated interface software using a readily and commercially available cable. The communications link must allow for control programming and uploading/downloading of data as a serial communication.
 - Ports: RS232, USB PC

– Remote Communication:

The control digital SCADA interface shall have these features.

- Two permanent communications ports for use with optional digital communication interfaces.
- Ports:
 - Ethernet 10BASE-T/100BASE-TX (Copper) RJ45
- The communications interfaces must be modular, user installable, and programmable. Permanently installed communications interfaces that are not user interchangeable are not acceptable.
- The following resident communications protocols, user-configurable and available for both communications ports shall be available:
 - IEC 60870-5-101/104 as mandatory
 - DNP3 (Serial and Ethernet)
 - MODBUS TCP/IP

2. Control Cyber Security

- The control access shall be protected by a configurable password.
- Password levels shall be available such as (Viewer, Changing parameter and configurations, and Full access).
- The communication module inside the control unit should be NTRA approved.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

3. Control Operational Requirements

- Software: A single software program must be available to make available the follow functionality:
 - Provide password protection to limit access to connect to the control
 - Display all settings, metering and control data
 - Fully program and configuring the control for operations and SCADA communications
 - Create download files which include of all settings, metering and control data
 - Create settings files for later uploading into a control
 - The latest version of the software must be fully compatible with all previous versions of control FW.
 - The required use of multiple software programs to provide the listed functionality is not acceptable.
- Tap changer operation: The control operation of the voltage regulator tap changer must:
 - Be fully configured to properly operate the tap-changer when a tap changer type selection is made.–The control must also be tested to confirm operational compatibility with the tap changer options available on the control.
 - Provide for all available tap changer standard options.
 - Provide for a user-defined tap changer option to operate tap changers not available as a standard option. The settings must be configured when the appropriate tap-changer type selection is made. The compatibility should not require fine tuning manual adjustments.
- Tap position indication: The control shall be capable all the following strategies to track and display an accurate tap position:
 - The control must be able to track tap position using the input of and operations counter switch
 - Use the neutral switch inputs and extreme tap positions (16 raise and 16 lower) to check the accuracy of tracking and corrects the indication in those positions if an error is found. When corrections are made, a counter shall record each of these occurrences.
 - Tap changer position correction function shall be available.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- Source bushing voltage: The control shall be capable of all the following strategies to display an accurate source bushing voltage:
 - Calculated based upon the type of voltage regulator, the displayed tap position, and, if available to the control, detailed apparatus design information. This measurement must be accurate within 1.5%.
 - Measured using a voltage measurement between the source and source-load bushings accurate within 1%.
- Load bushing voltage: It is not acceptable for the control to determine the load bushing voltage using a calculation based on the displayed tap position.
- Load current: The load current shall be displayed using a current transformer input with a nominal secondary current of 0.2 amps and be accurate within 1%.
- Time and Date display: The time and date displays shall have these features:
 - Time displayed in either of two user defined formats, 12-hour AM/PM or 24-hour.
 - Daylight savings time configuration setting.
 - Time setting using the control HMI, software interface or computer time sync using a software interface.
 - Date displayed in either of three user defined formats, MM/DD/YYYY, DD/MM/YYYY, YYYY/MM/DD.
- Tap changer operations counter: The control shall have the following operations counter features:
 - A six-digit electronic operations counter which is counts each tap change and that is user-settable when retrofitting the control. When the counter is set by a user, it will record the date, time and number of operations when set.
 - These additional operations counters with date and time stamp of last reset shall be available in the software:
 - Past 24 hours
 - Past 30 days
 - Operations since start of month
 - Last month

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- Operations since start of year
 - Last year
- Metering: The control shall provide a digital metering package with Class 1 accuracy which shall possess the following feature:
- Instantaneous metering data including: tap position, percent regulation, load voltage, source voltage, compensated voltage, load current, load current real, load current reactive, phase angle, power factor, kVA load, kW load, kvar load, line frequency, motor voltage, S-bushing voltage, L-bushing voltage, forward and reverse kW-h, and forward and reverse kvar-h.
 - The metering data shall include load voltage, source voltage, compensated voltage, load current, kVA load, power factor at kVA load, kW load, kvar load, tap position, and percent regulation.
 - Instantaneous metering to provide load and source voltages and phase angles for delta connected voltage regulators when using a multi-phase voltage regulator control. These metering values shall be incorporated into voltage regulation operations to minimize tapping operations and improve the accuracy of reverse voltage regulation.
 - Current and voltage harmonics to provide THD.
 - Fault detection to record the occurrence and duration of one levels of current faults.
- Data Acquisition: These control data recording features shall be available:
- A feature to compile date and time-stamped records of user-definable voltage regulator and control events. The 50 most recent events shall be viewable through the control HMI.
 - A data profiler feature to compile date and time-stamped metering data and settings. The data profiler shall also include this additional functionality:
 - Data recording start and stop date
 - Sampling interval
- Alarms: The control shall provide for these alarm features:
- Status alarms which become active when control and voltage regulator statuses change
 - Viewing of acknowledged and unacknowledged alarms on the control HMI or using the interface software

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

- Programmable supplementary current feature: The control shall have a feature which limits the range of regulation of the voltage regulator in a fashion similar to mechanical position indicator limit switches. When enabled, these functions shall automatically apply the soft limit setting based on control status, conditions, or metering points. An adaptive version of this feature shall also be available which will be activated based upon the load current. The Soft supplementary current feature increases the load current capacity of the voltage regulator by restricting the range of regulation.
- Alternate Configuration: Control shall have the ability to utilize three sets of alternate configuration settings. Three methods of activation of the alternate settings shall be available: locally using control HMI, remotely using SCADA communications, triggered using configurable logic equations. These settings shall be available, at a minimum, as alternate settings:
 - Forward and reverse basic control settings
- Leader/Follower Scheme: When utilizing single-phase controls to operate voltages regulators, the controls shall have a feature designed to keep two or three connected voltage regulators on the same mechanical tap position. The controls will utilize a fiber-optic communication loop to coordinate tap change operations and provide positive feedback in maintaining equal tap positions.
- Calibration: A calibration shall be available to calibrate the control for voltage and current.
 - Substation battery power that will power the control for normal operation from a substation battery. The substation battery may range of voltage from 24 to 125 Vdc.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 12-401-3
Step Voltage Regulators for MV Networks up to 24 kV	01-07-2025

4. Diagnostics and Maintenance

- Motor Trouble Status Alarm: The control shall be configurable to provide a status alarm to alert a user of tap-changer motor trouble. This status is to indicate an inability of the tap-changer motor to perform or complete an operation to manufacturer standards. The control must be able to report this status using a warning LED on the control.